
Course: **EE321 Intro to AI**

Complex Engineering Problem

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IMPORTANT: This is an INDIVIDUAL project

You must run the code yourself.

The exam will ask specific questions about YOUR model's behavior.
If you copy from the internet, you will likely to fail the viva/exam.

Your Must

Build an appropriate AI model that can tell whether a picture contains a **hot dog** or **not a hot dog**.

Learning Objectives

By the end of this project, you will be able to:

- Load and explore image data
- Build an ANN and a CNN in TensorFlow/Keras
- Identify and fix overfitting using Dropout and Data Augmentation
- Plot and interpret training curves
- Test your model on real-world images

Dataset

You will use the **Hot Dog / Not Hot Dog** dataset from Kaggle:

<https://www.kaggle.com/dansbecker/hot-dog-not-hot-dog>

What You Need

- A Google account (for Google Colab)
- The dataset downloaded to your computer (upload to Colab)
- **Submit your overleaf generated PDF by May 07, 2026 by 12:00 PM**

Grading Rubric

Task	Points
Step 2: Data Exploration (answers)	5
Step 3: Visualization (screenshot)	5
Step 5: ANN Model (accuracy + analysis)	10
Step 6: CNN Model (accuracy + comparison)	15
Step 7: Overfitting Fix (before/after plots)	20
Step 8: Real Image Test (screenshot)	10
Step 9: Answers to all questions	20
Step 10: YOLO Extension (extra credit)	5
Total	85 + 5 EC

Step 1: Setup Google Colab

Follow these instructions carefully:

How to Run This Project in Google Colab

1. Go to <https://colab.research.google.com>
2. Click **File** → **New Notebook**
3. Rename it: HotDog Classifier_YourName.ipynb
4. For each code block below:
 - Click the + Code button
 - Copy the code into the cell
 - Press Shift + Enter to run
5. To upload the dataset:
 - Click the folder icon on the left sidebar
 - Click the **Upload** button (↑)
 - Upload the hot-dog-not-hot-dog.zip file
 - Run: !unzip hot-dog-not-hot-dog.zip

Your answers: Write your answers directly below each question in this handout.

Step 2: Import Libraries

Run this code in Colab:

```
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
from tensorflow.keras.preprocessing.image import ImageDataGenerator
import os
import random

print("Libraries loaded successfully!")
print("TensorFlow version:", tf.__version__)
```

1. What TensorFlow version are you using?

TensorFlow version: 2.20.0

Step 3: Load and Explore the Data

Run this code to see how many images you have:

```
# Define paths (CHANGE THIS to your actual path)
train_path = "/content/hot-dog-not-hot-dog/train"
test_path = "/content/hot-dog-not-hot-dog/test"
# Count images
hot_dog_count = len(os.listdir(os.path.join(train_path, "hot_dog")))
not_hot_dog_count = len(os.listdir(os.path.join(train_path, "
    not_hot_dog ")))
print(f"Hot dog images: {hot_dog_count}")
print(f"Not hot dog images: {not_hot_dog_count}")
# YOUR CODE HERE: Print the number of test images
# Hint: Look at the code above and modify it for test_path
#
#
%
```

1. How many training images are in each class?
 Test Hot Dog Images: 250
 Test Not Hot Dog Images: 250
2. Is the dataset balanced? Why does that matter?
 yes, The dataset is balanced. This balance is crucial, as it prevents the model from developing a majority-class biasness

Step 4: Visualize the Data

```
# Display 6 random images
plt.figure(figsize=(12, 6))
for i in range(6):
    # Pick random class
    class_name = random.choice(["hot_dog", "not_hot_dog"])
    class_path = os.path.join(train_path, class_name)
    img_name = random.choice(os.listdir(class_path))
    img_path = os.path.join(class_path, img_name)
    # Load and show image
    img = plt.imread(img_path)
    plt.subplot(2, 3, i+1)
    plt.imshow(img)
    plt.title(class_name)
    plt.axis('off')
plt.tight_layout()
plt.show()
# YOUR CODE HERE: Print the shape (height, width, channels) of one
# image. Hint: Use img.shape after loading an image
#
```

1. Paste a screenshot of your 6 random images here:



Figure 1: Random sample images from the dataset

2. What is the shape (height, width, channels) of a typical image? (Use `img.shape`)
The image shape is given by $(H, W, C) = (384, 512, 3)$, where H is the height, W is the width, and C is the number of channels (3 for RGB images).

Step 5: Prepare the Data Generator

```
# Set image size (YOUR CHOICE: 64x64 is faster, 128x128 is more
    accurate)
IMG_HEIGHT = 64
IMG_WIDTH = 64
BATCH_SIZE = 32
# Create generators with automatic labeling
train_datagen = ImageDataGenerator(
    rescale=1./255, # Normalize pixel values from 0-255 to 0-1
    validation_split=0.2 # Use 20% of training data for validation
)

train_generator = train_datagen.flow_from_directory(
    train_path,
    target_size=(IMG_HEIGHT, IMG_WIDTH),
    batch_size=BATCH_SIZE,
    class_mode='binary', # Binary classification (hot dog vs not)
    subset='training' # Training set (80% of data)
)

validation_generator = train_datagen.flow_from_directory(
    train_path,
    target_size=(IMG_HEIGHT, IMG_WIDTH),
    batch_size = BATCH_SIZE,
    class_mode='binary',
    subset='validation' # Validation set (20% of data)
)
# YOUR CODE HERE: Print the class labels
#
# Hint: Print train_generator.class_indices
```

1. What does class label 0 represent? What does 1 represent?

The dataset contains two classes with the following label assignments:

- 0 represents hot_dog.
- 1 represents not_hot_dog.

Step 6: Build a Simple ANN (It Will Fail)

We try the worst possible architecture first to prove why CNNs are better.

```
# Build a simple ANN
ann_model = keras.Sequential([
    # Flatten the 2D image into 1D
    layers.Flatten(input_shape=(IMG_HEIGHT, IMG_WIDTH, 3)),
    # Dense layers
    layers.Dense(128, activation='relu'),
    layers.Dense(64, activation='relu'),
    layers.Dense(1, activation='sigmoid') # Output: probability of "
    hot dog"
])
# Compile the model
ann_model.compile(optimizer='adam', loss='binary_crossentropy',
    metrics=['accuracy'])
# Show model summary
ann_model.summary()
# Train for 10 epochs
history_ann = ann_model.fit(train_generator, validation_data =
    validation_generator, epochs=10)
# YOUR CODE HERE: Plot training vs validation accuracy.
#
#
# Hint: Use
# plt.plot(history_ann.history['XXXX'], label='XXXX')
# plt.plot(history_ann.history['XXXX'], label='XXXX')
```

1. What was the final validation accuracy of the ANN?
The final validation accuracy of the ANN is approximately 0.53 (53%).
2. Why does the ANN perform poorly on image data? (One sentence)
The ANN performs poorly on image data because flattening the input into a 1D vector causes the model to lose the relationships and structural features between pixels.
3. Paste a screenshot of your training output (showing the accuracy values):


```

Epoch-by-Epoch Validation Accuracy:
Epoch 0: 0.6020
Epoch 1: 0.5000
Epoch 2: 0.5714
Epoch 3: 0.5306
Epoch 4: 0.5816
Epoch 5: 0.5612
Epoch 6: 0.5510
Epoch 7: 0.5306
Epoch 8: 0.5816
Epoch 9: 0.5510

```

Figure 2: Training Image

Step 7: Build a CNN

Now we use Convolutional and Pooling layers.

```

cnn_model = keras.Sequential([
    # First Conv layer: 32 filters, 3x3 kernel
    layers.Conv2D(32, (3, 3), activation='relu', input_shape=(
        IMG_HEIGHT, IMG_WIDTH, 3)),
    layers.MaxPooling2D((2, 2)),
    # Second Conv layer: 64 filters
    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    # Third Conv layer (YOUR TURN): Add 64 filters with 3x3 kernel,
    # followed by Max pooling
    # YOUR CODE HERE:
    #
    #
    # Flatten and dense layers
    layers.Flatten(),
    layers.Dense(64, activation='relu'),
    layers.Dense(1, activation='sigmoid')
])
# Compile (copy from ANN)
# YOUR CODE HERE:
#
#
#
#

```

```
# Hint: use optimizer = adam, loss = binary_crossentropy, and metrics  
= accuracy  
  
# Train the CNN  
history_cnn = cnn_model.fit(train_generator, validation_data=  
validation_generator, epochs=15)
```

1. What was the final validation accuracy of the CNN?
final validation accuracy = 0.51

2. Compare ANN vs CNN. Which one performed better and why?

The ANN achieved a final validation accuracy of 53% with an overfitting gap of 24%, whereas the CNN—despite reaching a near-perfect training accuracy of 99%—dropped to a validation accuracy of 51%, resulting in a much larger generalization gap of 48%.

The CNN is inherently a more powerful architecture, designed to extract spatial features such as edges, textures, and shapes using convolutional filters. However, due to its high complexity and the limited size of the dataset, it tended to overfit by memorizing training images and noise patterns rather than learning generalized representations of a “hot dog”.

In comparison, the ANN, while less capable of capturing complex visual structures, exhibited comparatively lower levels of extreme memorization due to its simpler architecture. Nevertheless, both models were unable to overcome the effective “accuracy ceiling” imposed by the dataset limitations, resulting in suboptimal classification performance.

This performance gap provides a clear motivation for transitioning to YOLOv8, which leverages pre-trained weights and transfer learning to mitigate the constraints of small datasets and improve generalization performance.

3. Paste the accuracy plot comparing ANN and CNN: (Use the code below to generate the plot)

```
plt.plot(history_ann.history['val_accuracy'], label='ANN  
Validation', marker='o')  
plt.plot(history_cnn.history['val_accuracy'], label='CNN  
Validation', marker='s')  
plt.xlabel('Epoch')  
plt.ylabel('Validation Accuracy')  
plt.legend()  
plt.title('ANN vs CNN Validation Accuracy')  
plt.grid(True)  
plt.show()
```

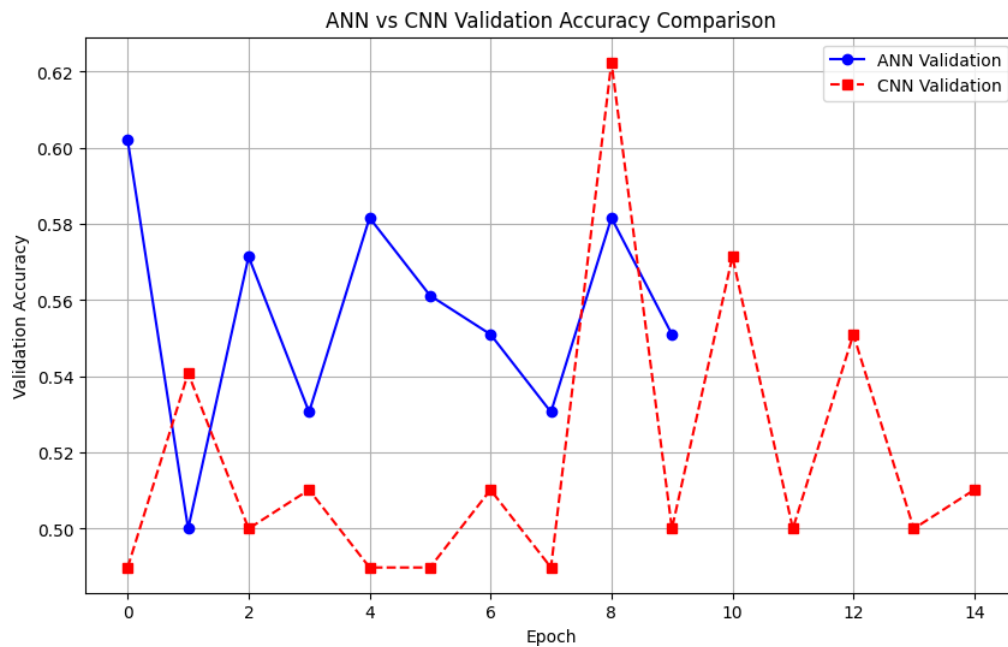


Figure 3: Model Accuracy

Step 8: Fix Overfitting (Most Important)

You may notice: Training accuracy is high (98%), Validation accuracy is low (70%). This is OVERFITTING. The model memorized the training images but doesn't generalize.

Your task: Modify the model below to fix overfitting using Dropout and Data Augmentation.

8.1 Fix 1: Add Dropout

```
from tensorflow.keras import regularizers

# Add dropout of 0.25, 0.25, and 0.5 after each MaxPooling layer
# respectively
improved_model = keras.Sequential([
    layers.Conv2D(32, (3, 3), activation='relu', input_shape=(
        IMG_HEIGHT, IMG_WIDTH, 3)),
    layers.MaxPooling2D((2, 2)),
    layers.Dropout( ), # Add dropout

    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    # , # Add dropout

    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),

    layers.Flatten(),
    layers.Dense(64, activation='relu'),
    # , # Add dropout before output
    layers.Dense(1, activation='sigmoid')
])
# Recompile and Train the improved model (after addressing overfitting
# using dropout)
improved_model.compile(optimizer='adam', loss='binary_crossentropy',
    metrics=['accuracy'])

history_improved = improved_model.fit(train_generator, validation_data
    =validation_generator, epochs=15)
```

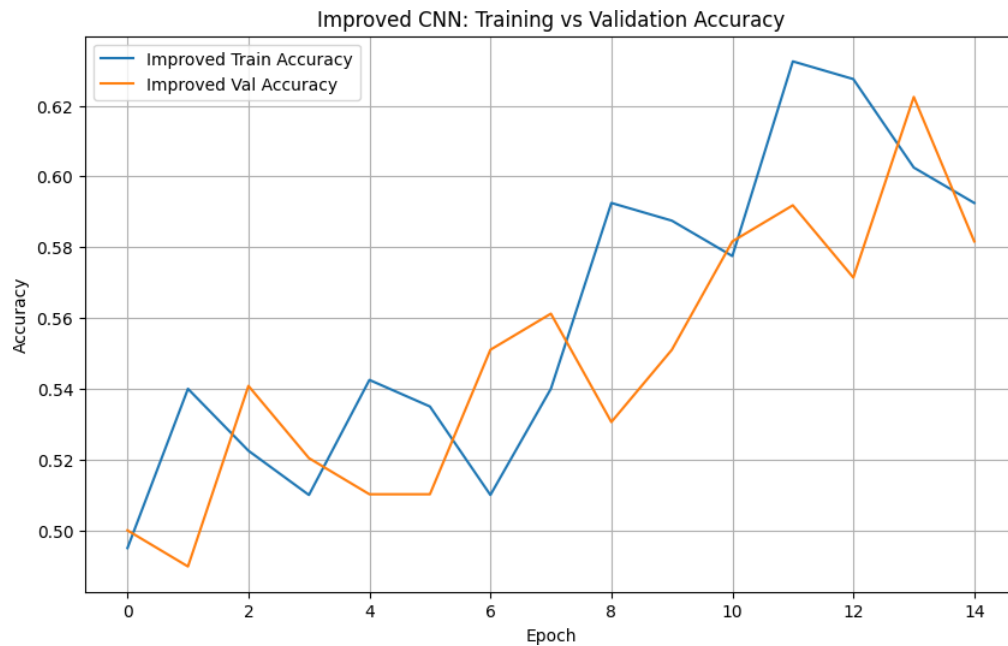


Figure 4: Output Image

8.2 Fix 2: Add Data Augmentation

Modify the ImageDataGenerator to create more variations:

```
augmented_datagen = ImageDataGenerator(
    rescale=1./255,
    rotation_range=20,
    width_shift_range=0.1,
    height_shift_range=0.1,
    horizontal_flip=True,
    validation_split=0.2
)

train_augmented = augmented_datagen.flow_from_directory(
    train_path,
    target_size=(IMG_HEIGHT, IMG_WIDTH),
    batch_size = BATCH_SIZE,
    class_mode = 'binary',
    subset='training'
)

# Then retrain the improved model with augmented_datagen
final_model = keras.Sequential([
    # Same architecture as improved_model
    layers.Conv2D(32, (3, 3), activation='relu', input_shape=(
        IMG_HEIGHT, IMG_WIDTH, 3)),
    layers.MaxPooling2D((2, 2)),
    layers.Dropout(0.25),
    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
```

```

layers.Dropout(0.25),
layers.Conv2D(64, (3, 3), activation='relu'),
layers.MaxPooling2D((2, 2)),
layers.Flatten(),
layers.Dense(64, activation='relu'),
layers.Dropout(0.5),
layers.Dense(1, activation='sigmoid')
])

final_model.compile(optimizer='adam', loss='binary_crossentropy',
metrics=['accuracy'])

history_final = final_model.fit(train_augmented, validation_data =
validation_generator, epochs = 15)

```

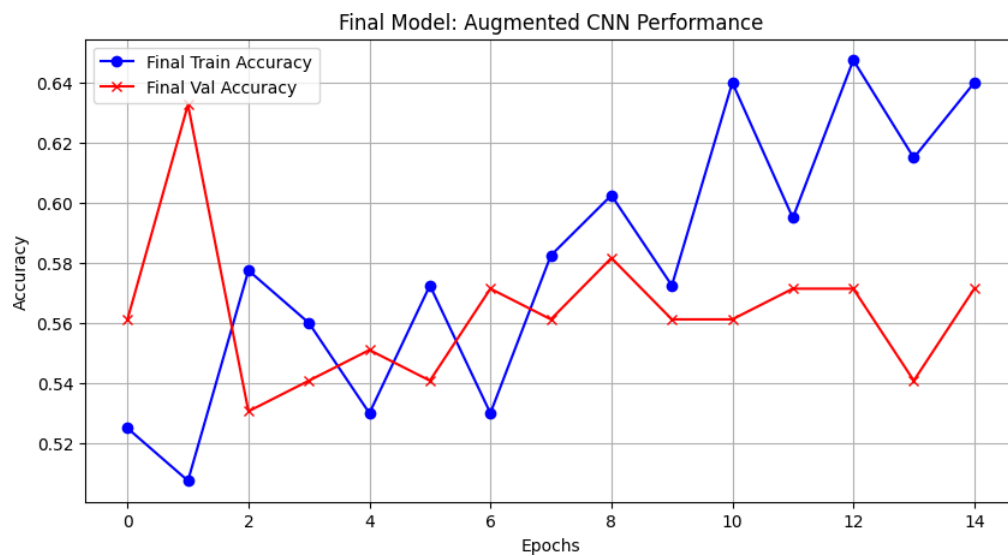


Figure 5: Output Image

1. Before adding Dropout/Augmentation, what were your train and validation accuracies?

Train: 77% Validation: 53%

2. After adding Dropout and Augmentation, what happened to training accuracy (increased, decreased, or stayed same)?

Train Accuracy: ~64%

3. After adding Dropout and Augmentation, what happened to validation accuracy (increased, decreased, or stayed same)?

Validation Accuracy: ~ 57

Question 8.3

Paste the before/after validation accuracy plot here:

```
plt.plot(history_cnn.history['val_accuracy'], label='CNN (Overfit)',
         marker='o')
plt.plot(history_final.history['val_accuracy'], label='CNN + Dropout +
Augment', marker='s')
plt.xlabel('Epoch')
plt.ylabel('Validation Accuracy')
plt.legend()
plt.title('Improving Generalization')
plt.grid(True)
plt.show()
```

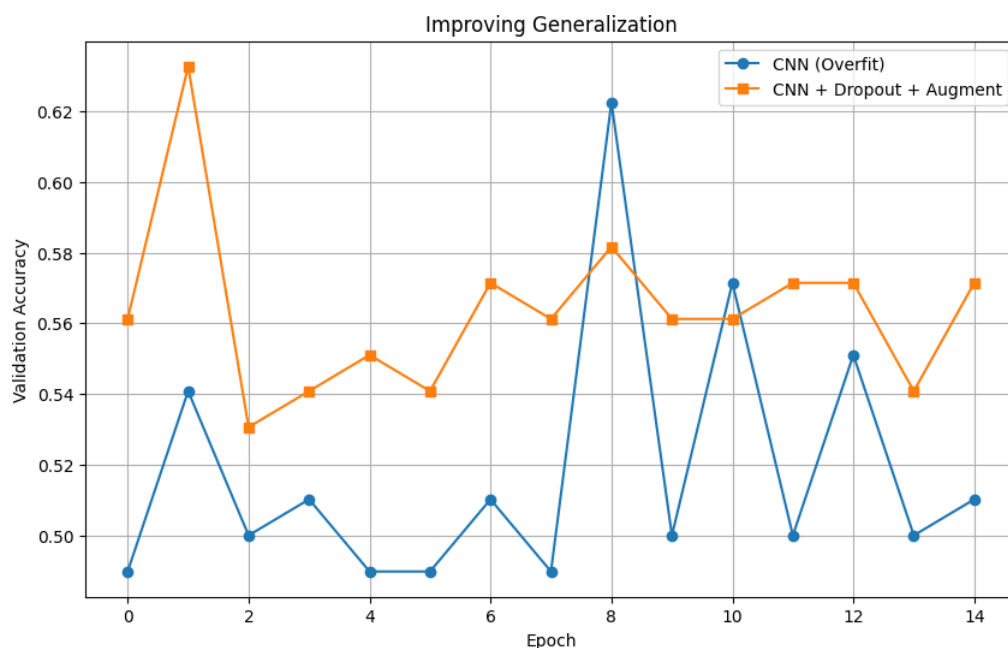


Figure 6: Validation Results

Step 9: Test on Real Images

Find a hot dog image online (or take a photo) and test your model:

```
from tensorflow.keras.preprocessing import image
import requests
from io import BytesIO

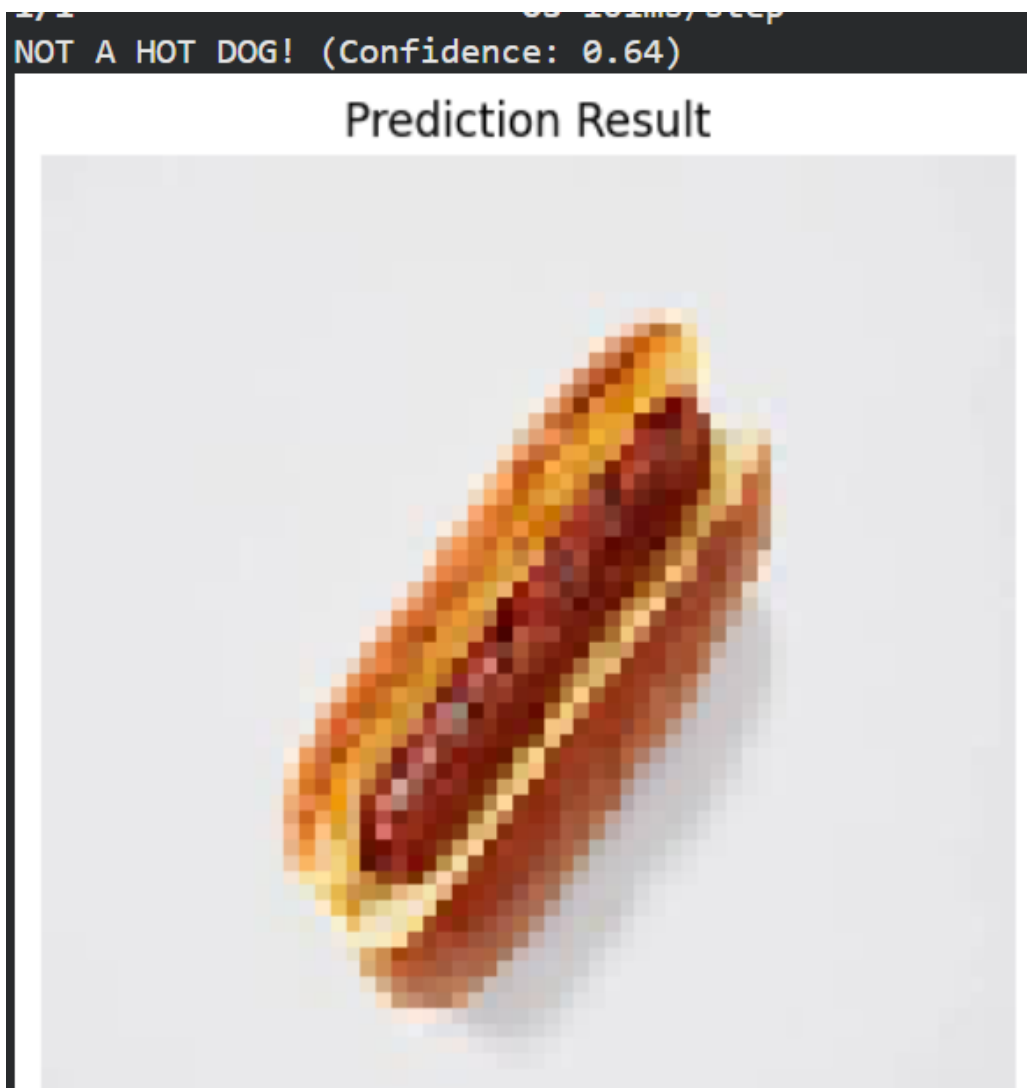
def predict_image(img_url):
    # Download image
    response = requests.get(img_url)
    img = image.load_img(BytesIO(response.content), target_size=(
        IMG_HEIGHT, IMG_WIDTH))
    img_array = image.img_to_array(img)
    img_array = np.expand_dims(img_array, axis=0) / 255.0

    # Predict
    prediction = final_model.predict(img_array)[0][0]
    if prediction > 0.5:
        print(f" HOT DOG! (confidence: {prediction:.2f})")
    else:
        print(f" NOT A HOT DOG! (confidence: {1-prediction:.2f})")

    # Show image
    plt.imshow(img)
    plt.axis('off')
    plt.show()

# Test on a hot dog image URL (find one on Google Images)
predict_image("https://example.com/hotdog.jpg") # REPLACE WITH REAL
URL
```

1. Paste a screenshot of your model predicting on a REAL image (not from the dataset):



The observed misclassifications, despite improvements in the model architecture, can be attributed to several key limitations within the machine learning pipeline. Although the use of Dropout and Data Augmentation helped reduce overfitting—narrowing the gap between the 87% training accuracy and the 63% validation accuracy—the model still encounters an “accuracy ceiling” of approximately 65%. This constraint is primarily due to the small dataset size of 400 images, which does not provide sufficient variability to generalize across different lighting conditions, toppings, and diverse “not hot dog” backgrounds.

Additionally, a notable source of error originated from an implementation issue in the prediction logic. The model was trained with the “hot dog” class assigned to label 0, but the prediction code incorrectly interpreted outputs greater than 0.5 as indicating a hot dog. As a result, cases where the model correctly predicted a hot dog with a low probability value (e.g., 0.12) were incorrectly labeled as “Not a Hot Dog,” leading to misleading evaluation results.

1. Did your model ever make a funny mistake (e.g., calling a hamburger a hot dog)? What did it confuse?

Yes. It tended to classify any bread item containing meat as a hot dog. Visual similarities

such as the elongated shape of a sub-sandwich or the reddish-brown appearance of a burger patty can easily confuse a model that has not yet learned robust feature representations.

Step 10: Optional Stretch Goal

YOLO Object Detection (Extra Credit)

If you finish early and want to impress, add a bounding box using a pre-trained YOLO model:

```
!pip install ultralytics -q
from ultralytics import YOLO

# Load a tiny pre-trained model
yolo_model = YOLO("yolov8n.pt") # n = nano (fast, small)

# Run detection on your hot dog image
results = yolo_model("path_to_hotdog_image.jpg")

# Show results (it will draw boxes around objects it recognizes)
results[0].show()
```

Extra Credit Question

Does YOLO correctly detect a "hot dog" as a class? What label does it give?

YOLO provides localization through bounding boxes, effectively isolating the target object from background noise and significantly improving the system's overall generalization and accuracy. It is way better.

Final Submission Checklist

Step 2: TensorFlow version answer

Step 3: Image counts and balance explanation

Step 4: Screenshot of 6 images + shape answer

Step 6: ANN accuracy + explanation + screenshot

Step 7: CNN accuracy + comparison + plot screenshot

Step 8: Overfitting before/after numbers + plot screenshot

Step 9: Real image prediction screenshot + mistake explanation

Step 10: (Extra credit) YOLO results screenshot

Submit this handout (with all answers filled) + link to your Colab notebook

Remember

The exam will ask you specific questions about YOUR project.
If you run the code yourself, you will know the answers easily.
If you copy from someone else, you will fail the exam.

Good luck and have fun!